

Lecture 5

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- Rates of convergence for matrix sequences
- Continuity in $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$
- Examples of continuous functions in higher dimensions

- Open subset
- Compact set
- Connected set
- Interior of a set
- Closure of a set

Rates of Convergence Revisited

The definitions for the orders of convergence of scalar sequences extend verbatim to sequences of matrices in $\mathbb{R}^{m \times n}$, with absolute value replaced by norm.

Definition

Let $(X_k)_{k=1}^{\infty}$ be a sequence in $\mathbb{R}^{m \times n}$ converging to X_* . We say $(X_k)_{k=1}^{\infty}$:

Converges:

converges:

$$\text{if } \lim_{k \rightarrow \infty} \|X_k - X_*\| = 0$$

linearly:

if there is some $N \in \mathbb{N}$ and $0 < c < 1$ such that $k > N$ implies

$$\|X_{k+1} - X_*\| \leq c \|X_k - X_*\|$$

superlinearly:

if there is some $N \in \mathbb{N}$ and $0 < c_k < 1$ such that $k > N$ implies

$$\|X_{k+1} - X_*\| \leq c_k \|X_k - X_*\| \text{ where } \lim_{k \rightarrow \infty} c_k = 0$$

sublinearly:

if there is some $N \in \mathbb{N}$ and $0 < c_k < 1$ such that $k > N$ implies

We can rewrite all of these definitions in terms of the ratio of the error at step $k + 1$ to the error at step k , raised to some power:

linearly:

if there is some $N \in \mathbb{N}$ and $0 < c < 1$ such that $k > N$ implies

$$\frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|} \leq c$$

superlinearly:

if there is some $N \in \mathbb{N}$ and $0 < c_k < 1$ such that $k > N$ implies

$$\frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|} \leq c_k \quad \text{where} \quad \lim_{k \rightarrow \infty} c_k = 0$$

sublinearly:

if there is some $N \in \mathbb{N}$ and $0 < c_k < 1$ such that $k > N$ implies

$$\frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|} \leq c_k \quad \text{where} \quad \lim_{k \rightarrow \infty} c_k = 1$$

quadratically:

if there is some $N \in \mathbb{N}$ and $c > 0$ such that $k > N$ implies

$$\frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|^2} \leq c$$

We can categorize these convergence rates by the power of the error term in the denominator and by the limiting value of the ratio.

The power of the error term in the denominator is the **order of convergence**, and the limiting value of the ratio is the **rate of convergence**.

Convergence Type	Order	Rate
Sublinear	1	1
Linear	1	$c < 1$
Superlinear	1	0
Quadratic	2	$c > 0$

To find the order and rate of convergence for a sequence, we need to find the largest possible power of the error term in the denominator that makes the ratio approach a constant.

Consider three sequences converging to 0:

1. $x_k = \frac{1}{2^k}$ (linear convergence)
2. $y_k = \frac{1}{k^2}$ (sublinear convergence)
3. $z_k = \frac{1}{2^{2k}}$ (quadratic convergence)

For the first sequence:

$$\frac{\text{error}_{k+1}}{\text{error}_k^1} = \frac{x_k - 0}{x_{k+1} - 0} = \frac{1/2^{k+1}}{1/2^k} = \frac{1}{2}$$

The ratio approaches a constant less than 1. This shows that:

- Order = 1 (we're raising the error to power 1)
- Rate = $\frac{1}{2}$ (the limiting value of the ratio)

Notice that this sequence does *not* converge quadratically. We can check this by calculating the ratio of the error at step $k + 1$ to the error at step k , raised to the power 2:

$$\frac{\text{error}_{k+1}}{\text{error}_k^2} = \frac{1/2^{k+1}}{(1/2^k)^2} = \frac{1/2^{k+1}}{1/2^{2k}} = \frac{1}{2^{1-k}} = 2^{k-1}$$

For k large, this ratio grows without bound. There's definitely no constant c such that $2^{k-1} \leq c$ for all k .

So to find the order of convergence, we need to find the power of the error term in the denominator that makes the ratio approach a constant.

For the second sequence:

$$\frac{y_{k+1}}{y_k} = \frac{1/(k+1)^2}{1/k^2} = \frac{k^2}{(k+1)^2} \rightarrow 1$$

The ratio approaches 1. This shows that: - Order = 1 (we're raising the error to power 1) - Rate = 1 (the limiting value of the ratio)

For the third sequence, we can start again by raising the error to the power 1:

$$\frac{z_{k+1}}{z_k} = \frac{1/2^{2^{k+1}}}{1/2^{2^k}} = 2^{-2^k}$$

This ratio approaches 0, which means the sequence converges faster than linearly.

Let's see if we can do better, by raising the error to the power 2:

$$\frac{z_{k+1}}{z_k^2} = \frac{1/2^{2^{k+1}}}{(1/2^{2^k})^2} = \frac{1/2^{2^{k+1}}}{1/2^{2^{k+1}}} = 1$$

The ratio approaches a constant. This shows that:

- Order = 2 (we're raising the error to power 2)
- Rate = 1 (the limiting value of the ratio)

All three sequences converge to 0, but in different ways:

- Linear convergence: order 1, rate $\frac{1}{2} < 1$
- Sublinear convergence: order 1, rate 1
- Quadratic convergence: order 2, rate 1

It would be nice to have a way to describe rates and orders of convergence generally, beyond just linear, quadratic, etc.

The idea: find the power of the error term in the denominator that makes the ratio approach a constant.

This is the “Goldilocks” power!

$$\frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|^\alpha}$$

If α is too high, the denominator goes to zero too quickly and the fraction goes to infinity.

If α is too low, the denominator goes to zero too slowly and the fraction goes to zero.

At the just right power, the fraction approaches a constant.

To do this formally, we'll need to allow for cases where the limit of the ratio is infinite or does not exist.

(For instance, if the sequence oscillates, the limit of the ratio will not exist.)

In every case, though, the limit of the supremum will exist (although it might be infinite).

Building up to our formal definition:

We'll define a function $r(\alpha)$ that gives us the limit of the supremum of the ratio:

$$r(\alpha) = \limsup_{k \rightarrow \infty} \frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|^\alpha}$$

Then $r(\alpha)$ will be 0 if α is too large, infinite if α is too small, and some finite value if α is just right.

Our final and formal definition:

Let $(X_k)_{k=1}^{\infty}$ be a convergent sequence with $\lim_{k \rightarrow \infty} X_k = X_*$. Furthermore, suppose there is some $M \in \mathbb{N}$ such that $X_k \neq X_*$ for all $k > M$. For $\alpha > 0$, set

$$r(\alpha) = \limsup_{k \rightarrow \infty} \frac{\|X_{k+1} - X_*\|}{\|X_k - X_*\|^\alpha}$$

There is some $\alpha_0 > 0$ and $r_0 \geq 0$ finite such that

$$r(\alpha) = \begin{cases} 0 & \text{if } \alpha < \alpha_0 \\ r_0 & \text{if } \alpha = \alpha_0 \\ \infty & \text{if } \alpha > \alpha_0 \end{cases}$$

The rate of convergence is r_0 and the order of convergence is α_0 .

Continuity

Definition (Continuity of matrix functions)

A function $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$ is continuous at $X \in \mathbb{R}^{m \times n}$ if

$$\lim_{Y \rightarrow X} f(Y) = f(X)$$

i.e., for all $\varepsilon > 0$ there is some $\delta > 0$ such that

$$\|f(Y) - f(X)\| < \varepsilon$$

whenever $\|Y - X\| < \delta$.

Another way to think about continuity is in terms of sequences.

If we have a sequence X_k that converges to X , and f is continuous at X , then the sequence $f(X_k)$ will converge to $f(X)$.

In other words, f is continuous at $X \in \mathbb{R}^{m \times n}$ if and only if

$$\lim_{k \rightarrow \infty} f(X_k) = f(X)$$

whenever $\lim_{k \rightarrow \infty} X_k = X$.

Continuity on subsets is defined in the same way as continuity on the whole space.

We say f is continuous on an open subset $\Omega \subseteq \mathbb{R}^{m \times n}$ if and only if f is continuous at X for all $X \in \Omega$.

We say f is Lipschitz on Ω if there is a Lipschitz constant $\gamma > 0$ such that

$$\|f(Y) - f(X)\| \leq \gamma \|Y - X\|$$

for all $X, Y \in \Omega$.

Working with continuous functions directly from the definition is often not useful/easy.

For example, showing that $e^{\tan x^2}$ is continuous would be tedious from the definition.

But if we know rules for how continuity works for different combinations of functions, we can often use these rules to more easily show that a function is continuous.

The following theorem gives various constructions of continuous functions.

Theorem

Let $f, g : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$ be continuous. Then the following functions are also continuous:

- i. $X \mapsto \|f(X)\|$
- ii. $X \mapsto f(X)^\top g(X)$
- iii. $X \mapsto \alpha f(X) + \beta g(X)$
- iv. $X \mapsto f(X)g(X)$, *when dimensions are compatible*
- v. $X \mapsto f(X)^\top$
- vi. $X \mapsto g(f(X))$, *when the functions can be composed*

Matrix and vector functions are continuous if and only if each of their components are continuous.

Formally:

Theorem

A function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{bmatrix}$$

is continuous iff each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous for all $i = 1, \dots, m$.

Likewise a function

$$f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}, \quad X = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix} \mapsto \begin{bmatrix} f_{11}(X) & \cdots & f_{1q}(X) \\ \vdots & \ddots & \vdots \\ f_{p1}(X) & \cdots & f_{pq}(X) \end{bmatrix},$$

is continuous iff each $f_{ij} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is continuous for all $i = 1, \dots, p$ and $j = 1, \dots, q$.

There are alternative definitions of continuity that are purely in terms of open and closed sets. They could be useful in proving various things although we do not formally need them in this course.

Theorem (Topological characterizations of continuity)

Let $f : \mathcal{U} \rightarrow \mathcal{V}$. The following are equivalent:

- i. f is continuous;
- ii. $B \subseteq \mathcal{V}$ open $\Rightarrow f^{-1}(B) \subseteq \mathcal{U}$ open;
- iii. $C \subseteq \mathcal{V}$ closed $\Rightarrow f^{-1}(C) \subseteq \mathcal{U}$ closed;
- (iv) $B \subseteq \mathcal{V} \Rightarrow \overline{f^{-1}(B)} \subseteq f^{-1}(\bar{B})$;
- v. $A \subseteq \mathcal{U} \Rightarrow f(\bar{A}) \subseteq \overline{f(A)}$;
- vi. $B \subseteq \mathcal{V} \Rightarrow f^{-1}(\text{int}(B)) \subseteq \text{int}(f^{-1}(B))$.

Here $\text{int}(S)$ and $\bar{S}$ denote the interior and closure of the set S respectively. Interestingly, the statement

$$A \subseteq \mathbb{U} \Rightarrow \text{int}(f(A)) \subseteq f(\text{int}(A))$$

is neither implied by nor implies continuity of f .

Standard properties of continuous functions

Continuous functions preserve topological properties

The following are some standard properties of continuous functions. Feel free to quote and use them.

When going backwards, continuous functions preserve “openness” and “closedness” of sets.

That is, the preimage of an open set is open, and the preimage of a closed set is closed.

Theorem (Topological properties of continuity)

Let $f : \mathbb{U} \rightarrow \mathbb{V}$ be continuous. Then

- $f^{-1}(\text{open})$ is open;
- $f^{-1}(\text{closed})$ is closed;

When going forwards, continuous functions preserve “compactness” and “connectedness” of sets.

That is, the image of a compact set is compact, and the image of a connected set is connected.

Formally:

Theorem (Topological properties of continuity continued)

- f (compact) is compact;
- f (connected) is connected.

Example

The function $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ by $f(X) = X^2$ is continuous.

There are two ways to see this:

1. Use Theorem 1.2 observing that every entry in the matrix X^2 is a polynomial function (therefore continuous) in the entries of X , or
2. Directly using definition.

How would we show this directly from the definition in the univariate case?

Suppose we want to prove that $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$, is continuous at $y \in \mathbb{R}$.

Using the definition, we need to show that for any $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|x^2 - y^2| < \varepsilon$$

whenever $|x - y| < \delta$.

$$|x^2 - y^2| = |x - y||x + y| \leq |x - y|(|x| + |y|);$$

We need to provide a δ such that

$$|x^2 - y^2| < \varepsilon$$

whenever $|x - y| < \delta$.

Note that if we choose x such that $|x - y| \leq 1$, and so $|x| \leq 1 + |y|$, and so the above inequality becomes

$$|x^2 - y^2| \leq |x - y|(1 + 2|y|) < \varepsilon$$

$$|x^2 - y^2| \leq |x - y|(1 + 2|y|) < \varepsilon$$

What δ will make this true whenever $|x - y| < \delta$?

$$|x^2 - y^2| \leq |x - y|(1 + 2|y|) \leq \delta(1 + 2|y|) < \varepsilon$$

So we can choose $\delta = \min\{1, \varepsilon/(1 + 2|y|)\}$. This will ensure both that $|x - y| \leq 1$, which we used above, and that $|x^2 - y^2| < \varepsilon$.

If we try to do the same for matrix variables, we realize that it fails in the first step. The first step in the univariate case was

$$|x^2 - y^2| = |x - y||x + y| \leq |x - y|(|x| + |y|);$$

This relied on the fact that $x^2 - y^2 = (x - y)(x + y)$.

But with matrices, because matrix multiplication is not commutative, $XY \neq YX$ in general, so we have

$$(X - Y)(X + Y) = X^2 - Y^2 - XY + YX \neq X^2 - Y^2$$

So we can't do the same proof!

Instead, we'll do this slightly more involved proof:

$$\begin{aligned}X^2 - Y^2 &= (X - Y)(X + Y) - XY + YX = (X - Y)(X + Y) - XY + Y^2 - Y^2 + YX \\&= (X - Y)(X + Y) - (X - Y)Y + Y(X - Y)\end{aligned}$$

By triangle inequality and submultiplicativity,

$$\begin{aligned}\|X^2 - Y^2\| &\leq \|X - Y\|\|X + Y\| + \|X - Y\|\|Y\| + \|Y\|\|X - Y\| \\&\leq \|X - Y\|(\|X\| + 3\|Y\|)\end{aligned}$$

and if we choose X such that $\|X - Y\| \leq 1$, and so $\|X\| \leq 1 + \|Y\|$, we get

$$\leq \|X - Y\|(1 + 4\|Y\|) < \varepsilon$$

whenever $\|X - Y\| < \min\{1, \varepsilon/(1 + 4\|Y\|)\} =: \delta$.

Sometimes it's easier to use definition and sometimes it's easier to use Theorem 1.2.

Takeaway: the definition for continuity wasn't different for the matrix case, but we had to watch out because matrix multiplication is not commutative!

Example

Transposition $\mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{n \times m}, X \mapsto X^\top$, is continuous.

Why?

$$\|X^\top - Y^\top\| = \|(X - Y)^\top\| = \|X - Y\|$$

So for any $\varepsilon > 0$, we can choose say $\delta = \varepsilon/2$ to make $\|X^\top - Y^\top\| < \varepsilon$ whenever $\|X - Y\| < \delta$.

Example

Norm $\|\cdot\| : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}, X \mapsto \|X\|$, is continuous since

$$|\|X\| - \|Y\|| \leq \|X - Y\|.$$

In fact this shows that $\|\cdot\|$ is Lipschitz with Lipschitz constant 1. This example specializes to $\mathbb{R}^m$ by setting $n = 1$.

Example

Let $A \in \mathbb{R}^{p \times m}$ and $B \in \mathbb{R}^{n \times q}$. The function $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}, X \mapsto AXB$, is continuous since by submultiplicativity,

$$\|f(X) - f(Y)\| = \|AXB - AYB\| = \|A(X - Y)B\| \leq \|A\| \|X - Y\| \|B\|$$

In fact this shows that f is Lipschitz with Lipschitz constant $\|A\| \|B\|$. This example specializes to functions $\mathbb{R}^{m \times n} \rightarrow \mathbb{R}^p$ or $\mathbb{R}^m \rightarrow \mathbb{R}^p$ by setting $q = 1$ or $m = q = 1$.

Example

The determinant $\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ is continuous since it is a polynomial function in the entries of the matrix

$$\det(X) = \sum_{\sigma \in \mathfrak{S}_n} \left(\operatorname{sgn}(\sigma) \prod_{i=1}^n x_{i, \sigma_i} \right)$$

Example

The inverse operation on $\text{GL}(n) := \{X \in \mathbb{R}^{n \times n} : \det(X) \neq 0\}$, since by Cramer's Rule,

$$X^{-1} = \text{adj}(X) / \det(X)$$

where $\text{adj}(X)$ is the adjugate matrix of X . So every entry in the inverse matrix is a rational function of entries of the matrix, e.g.,

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{x_{22}}{x_{11}x_{22}-x_{12}x_{21}} & \frac{-x_{12}}{x_{11}x_{22}-x_{12}x_{21}} \\ \frac{-x_{21}}{x_{11}x_{22}-x_{12}x_{21}} & \frac{x_{11}}{x_{11}x_{22}-x_{12}x_{21}} \end{bmatrix},$$

and the denominators are nonzero. So $f : \text{GL}(n) \rightarrow \text{GL}(n), X \mapsto X^{-1}$ is continuous.

Coercive functions

In optimization, we often deal with unbounded functions or optimize over non-compact sets.

When can we guarantee a minimum?

All we need is that it has a **compact sublevel set**.

Definition (Sublevel set)

Let $f : \Omega \subseteq \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$. Then the set

$$E_\alpha = \{X \in \Omega : f(X) \leq \alpha\} = f^{-1}((-\infty, \alpha])$$

is called the α -sublevel set of f .

Example of a function that with an unbounded sublevel set

Example of a function with a non-closed sublevel set

Consider the function

$$f(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

The sublevel set $E_{0.5} = \{x \in \mathbb{R} : f(x) \leq 0.5\}$ is not closed because it does not contain the point $x = 0$ – even though this point can be approached arbitrarily closely by sequences of points in $E_{0.5}$.

So the set is not closed, and thus not compact.

What about other sublevel sets?

Every sublevel set for $\alpha \leq 1$ is not closed, for the same reason as before.

What about sets with $\alpha \geq 1$? **Every** point in our function is ≤ 1 .

For $\alpha \geq 1$, the sublevel set $E_\alpha = \{x \in \mathbb{R} : f(x) \leq \alpha\} = \mathbb{R}$.

$\mathbb{R}$ is closed, but it's not bounded. (We can't find a ball that contains the whole set.)

We've just shown that our function doesn't have compact sublevel sets for any $\alpha \geq 1$ nor for $\alpha \leq 1$.

This function does not have any compact sublevel sets.

Definition (Coercive function)

A function $f : \Omega \rightarrow \mathbb{R}$ is called coercive (or 0-coercive) if it has a compact sublevel set $E_\alpha \neq \emptyset$ for some $\alpha \in \mathbb{R}$.

So for our three examples above, only the first function was coercive.

- $f(x) = x^2 + y^2$ is coercive.
- $f(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ is not coercive because every sublevel set is unbounded.
- $f(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ is not coercive because every sublevel set is either unbounded or not closed.

Why do we care about coercive functions?

If a function is coercive, then it attains its minimum!

Theorem (Guaranteeing a minimizer)

If $f : \Omega \rightarrow \mathbb{R}$ is coercive and continuous, then it attains its minimum in Ω .

Proof. Assume E_α is not empty. Draw a picture to see that

$$\inf_{X \in \Omega} f(X) = \inf_{X \in E_\alpha} f(X).$$

Conclude from the extreme value theorem for compact sets. Details in Homework 3.

Intuitively, the distance from a point to a set is the smallest distance from the point to any point in the set.

Formally:

Example (Distance to a set)

Let $\emptyset \neq S \subseteq \mathbb{R}^{m \times n}$ and $\|\cdot\|$ be a norm on $\mathbb{R}^{m \times n}$. Define $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ by

$$f(X) = \inf_{Z \in S} \|X - Z\| = \text{dist}(X, S)$$

$$f(X) = \inf_{Z \in S} \|X - Z\| = \text{dist}(X, S)$$

Let's show that f is Lipschitz.

By the triangle inequality,

$$\|X - Z\| \leq \|X - Y\| + \|Y - Z\|.$$

Taking infimum over all $Z \in S$,

$$f(X) \leq \|X - Y\| + f(Y)$$

$$f(X) - f(Y) \leq \|X - Y\|$$

$$f(X) \leq \|X - Y\| + f(Y)$$

$$f(X) - f(Y) \leq \|X - Y\|$$

Switching X and Y ,

$$f(Y) - f(X) \leq \|Y - X\| = \|X - Y\|.$$

Since we have shown both that $f(X) - f(Y) \leq \|X - Y\|$ and that $f(Y) - f(X) \leq \|X - Y\|$, it follows that

$$|f(X) - f(Y)| \leq \|X - Y\|$$

and so $f \in \text{Lip}_1(\mathbb{R}^{m \times n})$, i.e., Lipschitz with Lipschitz constant 1 .

If S is a bounded set, then our function f is a coercive function (it doesn't matter if S is closed or open or neither.)

Why?

Let's think about the sublevel set with $\alpha = 0$. If we can show that it's compact, then f is coercive and we're done.

The sublevel set $E_0 = \{X \in \mathbb{R}^{m \times n} : f(X) \leq 0\}$ is the set of all points that are at distance 0 from S .

This includes all points in S – but also all points that can be approached arbitrarily closely by points in S .

So if S is an open set, E_0 will contain S *plus its boundary*. E_0 will therefore be closed.

So if S is a bounded set, then E_0 is compact and f is coercive.

Formally, writing $\bar{S}$ as the closure of S (the set of all points in S plus its boundary),

$$\begin{aligned} E_0 &= \{X \in \mathbb{R}^{m \times n} : f(X) \leq 0\} \\ &= \left\{X \in \mathbb{R}^{m \times n} : \inf_{Z \in S} \|X - Z\| = 0\right\} = \bar{S} \end{aligned}$$

is closed and bounded, and thus compact.

Example: distance to a point

Let's consider the function that gives the distance from any point in a set to a fixed point within that set. Will this be a coercive function?

Let $X \in \mathbb{R}^{m \times n}$ be our fixed point.

Let S be a set in $\mathbb{R}^{m \times n}$.

Our function is $g(Z) = \|X - Z\|$.

In general, the function $g : S \rightarrow \mathbb{R}, Z \mapsto \|X - Z\|$ is not coercive if S is not closed.

However, if S is closed, g is coercive.

Why?

To show that g is coercive, we need to show that there is some α such that the α -sublevel set is compact – closed and bounded.

Take any $Y \in S$.

Let $\alpha = \|X - Y\|$. Then the α -sublevel set will be all points in S that are at distance $\leq \alpha$ from X – that is, points in S that are at least as close to X as Y is.

So $E_\alpha = \{Z \in S : \|X - Z\| \leq \alpha\} = S \cap \{Z \in \mathbb{R}^{m \times n} : \|X - Z\| \leq \alpha\}$.

$$E_\alpha = \{Z \in S : \|X - Z\| \leq \alpha\} = S \cap \{Z \in \mathbb{R}^{m \times n} : \|X - Z\| \leq \alpha\}$$

The set $\{Z \in \mathbb{R}^{m \times n} : \|X - Z\| \leq \alpha\}$ is a ball of radius α centered at X . This is a bounded set by definition since every point is at most distance α from X .

The intersection of a bounded set and any other set will still be bounded, so E_α is bounded.

Consider the closed set of real numbers $[0, \alpha]$.

Note that we could define Z as the set of all matrices Z such that the distance from X to Z is in this set:

$$Z = g^{-1}([0, \alpha]) = \{Z \in \mathbb{R}^{m \times n} : \|X - Z\| \in [0, \alpha]\}$$

In other words, Z is the *preimage* of the closed set $[0, \alpha]$ under the function g .

Since g is continuous, and the preimage of a closed set is closed, Z is closed.

Therefore, $E_\alpha = S \cap Z$ is closed (it's the intersection of two closed sets, and S was closed by assumption).

We have shown that E_α is closed and bounded, and thus compact. Therefore, g is coercive.

Derivatives in multivariate functions

We will start today by reviewing the basic types of derivatives for multivariate functions that you have already seen.

But a warning: the formulas we will see today will not be very useful for our purposes!

Prof. Lim's lecture notes call these "dumb formulas" because they are so useless in practice.

The simplest type of derivative for multivariate functions is the partial derivative.

Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, i.e.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix}$$

Then the i th partial derivative of f at $\mathbf{x}$, if it exists, is given by

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}.$$

If $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$, i.e.,

$$X = [x_{ij}]_{i,j=1}^{m,n} \mapsto [f_{kl}(X)]_{k,\ell=1}^{p,q}$$

then the (i, j) th partial derivatives are given by

$$\frac{\partial f}{\partial x_{ij}}(X) = \lim_{h \rightarrow 0} \frac{f(X + hE_{ij}) - f(X)}{h}.$$

If f is a vector-valued or matrix-valued function, then

$$\frac{\partial f}{\partial x_i} = \begin{bmatrix} \frac{\partial f_1}{\partial x_i}(\mathbf{x}) \\ \vdots \\ \frac{\partial f_m}{\partial x_i}(\mathbf{x}) \end{bmatrix}, \quad \frac{\partial f}{\partial x_{ij}}(X) = \left[\frac{\partial f_{k\ell}}{\partial x_{ij}}(X) \right]_{k,\ell=1}^{p,q}$$

Note that $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$ includes vector-valued function of matrix variables $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^p$ (set $q = 1$) and matrix-valued function of vector variables $f : \mathbb{R}^m \rightarrow \mathbb{R}^{p \times q}$ (set $n = 1$).

Partial derivatives are special cases of **directional derivative**: Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and fix $\mathbf{x} \in \mathbb{R}^n$. The directional derivative along $\mathbf{v}$, if it exists, is

$$\partial_{\mathbf{v}} f(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h}$$

Similarly for $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$ and $V \in \mathbb{R}^{m \times n}$:

$$\partial_V f(X) = \lim_{h \rightarrow 0} \frac{f(X + hV) - f(X)}{h}$$

Partial derivatives (and all directional derivatives) are all derivatives along a *line*. They are essentially univariate.

Their definitions are just minor variations of the univariate version.

The true multivariate derivative will require us to radically alter our idea of a derivative, namely,

A DERIVATIVE IS A LINEAR MAP

Let Ω be an open set in $\mathbb{R}^n$, and let $f : \Omega \rightarrow \mathbb{R}$ be a C^2 function on Ω .

Gradient, $\nabla f : \Omega \rightarrow \mathbb{R}^n$ given by

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^n$$

Hessian, $\nabla^2 f : \Omega \rightarrow \mathbb{R}^{n \times n}$ given by

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) \right]_{i,j=1}^n \in \mathbb{R}^{n \times n}$$

Note that this is a symmetric matrix, since

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

For a vector-valued function $f : \Omega \rightarrow \mathbb{R}^m$, we have the following generalization of gradient.

Jacobian, $Jf: \Omega \rightarrow \mathbb{R}^{m \times n}$ given by

$$Jf(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}) \end{bmatrix} = \left[\frac{\partial f_i}{\partial x_j}(\mathbf{x}) \right]_{i,j=1}^{m,n} \in \mathbb{R}^{m \times n}$$

If $m = 1$, i.e., $f : \Omega \rightarrow \mathbb{R}$ is scalar-valued, the Jacobian is related to the gradient in that

$$\nabla f(\mathbf{x}) = (Jf(\mathbf{x}))^\top$$

and it is related to the Hessian in that

$$\nabla^2 f(\mathbf{x}) = J\nabla f(\mathbf{x})$$

Gradient and Hessian are more important in optimization.

As we will see, a necessary condition for a local minimum at $\mathbf{x}$ is $\nabla f(\mathbf{x}) = \mathbf{0}$

A sufficient condition for a local minimum is $\nabla f(\mathbf{x}) = \mathbf{0}$ and $\nabla^2 f(\mathbf{x}) \succ 0$

(the shorthand ' $\succ 0$ ' means 'is positive definite.')

These formulas are far from adequate for various reasons:

- they are pretty much useless for computing gradients and Hessians of many important functions, e.g., $\det(X)$ or $\text{tr}(X^{-1})$ or $\log \det(X)$;
- even if you could use them, e.g., gradient of $\mathbf{x}^\top A \mathbf{x}$ or $\mathbf{a}^\top X \mathbf{b}$, there is a much neater and easier way to get the same result;

- We can't extend the formulas above to get the Hessian or Jacobian of a function of a matrix variable.

(We *can* do it for the gradient, though...)

h

$$\nabla f(X) = \begin{bmatrix} \frac{\partial f}{\partial x_{11}}(X) & \dots & \frac{\partial f}{\partial x_{1n}}(X) \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_{m1}}(X) & \dots & \frac{\partial f}{\partial x_{mn}}(X) \end{bmatrix} = \left[\frac{\partial f}{\partial x_{ij}}(X) \right]_{i,j=1}^{m,n} \in \mathbb{R}^{m \times n}$$

- if you have a matrix-valued function of a matrix $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$, say, $f(X) = X^2$, none of these formulas will allow you to find its first derivative;
- they do not generalize to higher-order derivatives.

- Multivariate convergence is much the same as univariate convergence, just using the norm
 - Formula for finding the order and rate of convergence
- Multivariate continuity is much the same as univariate continuity.
 - $\delta - \epsilon$ definition of continuity
 - Coordinatewise theorem
 - Function composition
- Coercive functions
- Multivariate differentiability review
 - Partial derivatives
 - Gradient
 - Hessian
 - Jacobian
 - Next lecture, better ways to calculate these!